DATA SHEET

MXG X-Series Signal Generator N5181B Analog & N5182B Vector

9 kHz to 3 or 6 GHz
9 kHz to 7.2 [1] GHz



1. Only applicable to N5182B + N5182BX07 Frequency Extender

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Definitions and Conditions

Specifications represent warranted performance of a calibrated instrument that has been stored for a minimum of 2 hours within the operating temperature of 0 to 55 °C, unless otherwise stated, and after a 45 minute warm-up period. The specifications include measurement uncertainty. Data represented in this document are specifications unless otherwise noted.

Typical (typ) describes additional product performance information that is not covered by the product warranty. It is performance beyond specifications that 80 percent of the units exhibit with a 90 percent confidence level at room temperature (approximately 25 °C). Typical performance does not include measurement uncertainty.

Nominal (nom) values indicate the expected mean or average performance, or an attribute whose performance is by design, such as the 50 ohm connector. This data is not warranted and is measured at room temperature (approximately 25 °C).

Measured (meas) describes an attribute measured during the design phase for purposes of communicating expected performance, such as amplitude drift vs. time. This data is not warranted and is measured at room temperature (approximately 25 °C).



Pure and Precise

On the path to better performance, the Keysight Technologies, Inc. MXG X-Series signal generators are fine-tuned to be your "golden transmitter" in R&D. Whether you're pushing for a linear RF chain or an optimized link budget, the analog and vector MXG models deliver what you need: phase noise, ACPR, channel coding, and more. Take your devices and designs to the limit with the MXG.

Frequency Specifications

Frequency range		
Frequency range	**Option 503**	**9 kHz (5 MHz I/Q mode) to 3 GHz**
	Option 506	9 kHz (5 MHz I/Q mode) to 6 GHz
	Option 506 + FRQ	9 kHz to 7.2 GHz [1]
Resolution	0.001 Hz	
Phase offset	Adjustable in nominal 0.1° increments	

Frequency bands [2]		
Band	**Frequency range**	**N**
1	9 kHz to < 5 MHz	1 (digital synthesis)
1	5 to < 250 MHz	1
2	250 to < 375 MHz	0.25
3	375 to < 750 MHz	0.5
4	750 to < 1500 MHz	1
5	1500 to < 3000.001 MHz	2
5	3000.001 to 6000 MHz	4

1. Only applicable to N5182B; requires option 506 and N5182BX07 Frequency Extender.
2. N is a factor used to help define certain specifications within the document.

Frequency switching speed [1, 2]			
	Standard	Option UNZ [3]	Option UNZ, typical
CW mode			
SCPI mode	≤ 5 ms, typical	≤ 1.15 ms	≤ 950 µs
List/step sweep mode	≤ 5 ms, typical	≤ 900 µs	≤ 800 µs
Digital modulation on (N5182B only)			
SCPI mode	≤ 5 ms, typical	≤ 1.15 ms	≤ 1.05 ms
List/step sweep mode	≤ 5 ms, typical	≤ 900 µs	≤ 800 µs

1. Time from receipt of SCPI command or trigger signal to within 0.1 ppm of final frequency or within 100 Hz, whichever is greater.
2. With internal channel corrections on, the frequency switching speed is <1.3 ms measured for list mode and SCPI mode cached frequency points. For the initial frequency point in SCPI mode the time is <3.3 ms measured. The instrument will automatically cache the most recently used 1024 frequencies. There is no speed degradation for amplitude-only changes.
3. Specifications apply when statues register updates are off. For export control purposes CW switching speed to within 0.05% of final frequency is 190 µs (measured).

Frequency reference	
Accuracy	**± (time since last adjustment x aging rate) ± temperature effects ± line voltage effects ± calibration accuracy**
Internal time base reference oscillator aging rate [1]	< ± 1 x 10-7/year < ± 5 x 10-10/day after 30 days
Initial achievable calibration accuracy	± 4 x 10-8 or ± 40 ppb
Adjustment resolution	< 1 x 10-10
Temperature effects	< ± 2 x 10-10, nominal
Line voltage effects	< ± 1 x 10-9 for ± 10% change, nominal

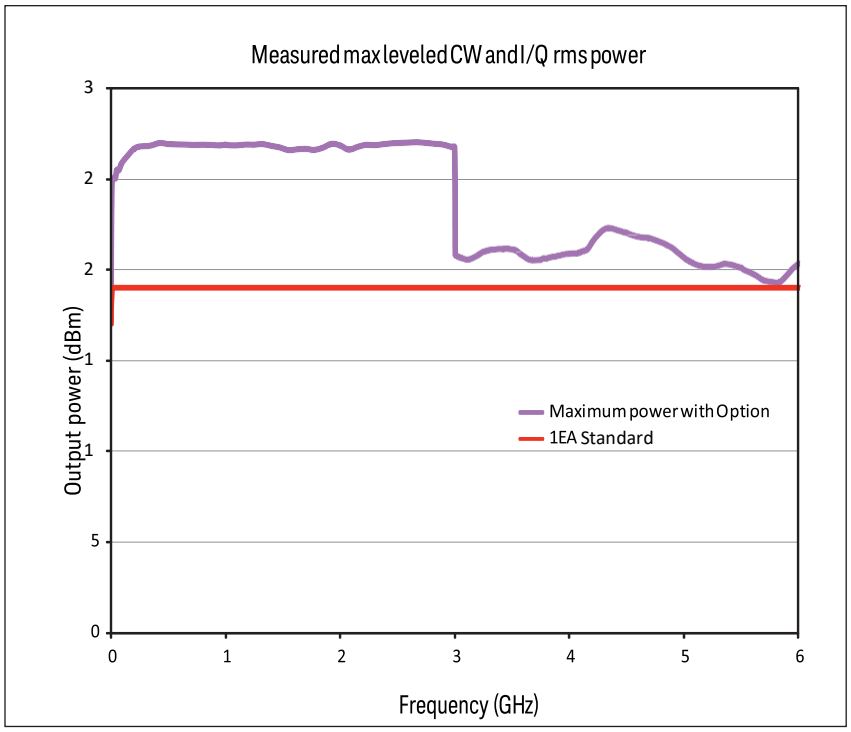
Reference output	
Frequency	10 MHz
Amplitude	≥ +4 dBm, nominal into 50 Ω load
External reference input	
Input frequency, standard	10 MHz
Input frequency, Option 1ER	1 to 50 MHz (in multiples of 0.1 Hz) [2]
Stability	Follows the stability of external reference input signal
Lock range	± 1 ppm
Amplitude	–3 dBm to +20 dBm, nominal
Impedance	50 Ω, nominal
Waveform	Sine or square
Sweep modes (frequency and amplitude)	
Operating modes	Step sweep (equally spaced frequency and amplitude or logarithmically spaced frequency steps) List sweep (arbitrary list of frequency and amplitude steps) Simultaneously sweep waveforms with N5182B; see Baseband Generator section for more detail
Sweep range	Within instrument frequency range
Dwell time	100 µs to 100 s
Number of points	2 to 65535 (step sweep) 1 to 3201 (list sweep)
Step change	Linear or logarithmic
Triggering	Free run, trigger key, external, timer, bus (GPIB, LAN, USB)

1. Not verified by Keysight N7800A TME Calibration and Adjustments Software. Daily aging rate may be verified as a supplementary chargeable service, on request.
2. Close-in phase noise will degrade when reference input is tuned away from 10 MHz.

Amplitude Specifications

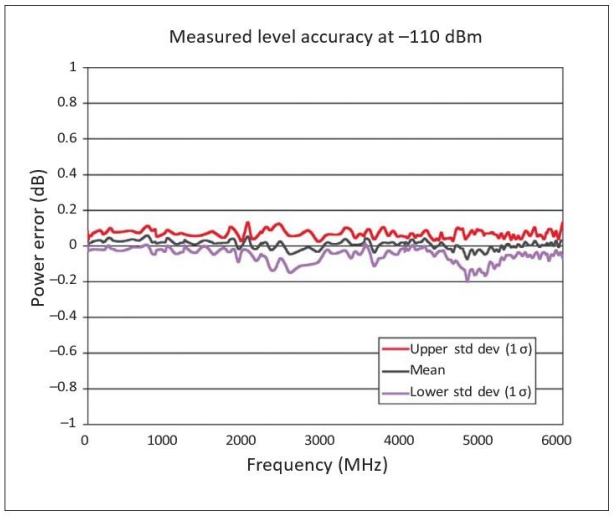
Output parameters		
Settable range	+19 to -144 dBm (Standard) +30 to -144 dBm (Option 1EA)	
Resolution	0.01 dB	
Step attenuator	0 to 130 dB in 5 dB steps electronic type	
Connector	Type N 50 Ω, nominal	
Max output power 1 () = typical		
Frequency	**Standard**	**Option 1EA**
9 kHz to 10 MHz	+13 dBm	+17 dBm (+18 dBm)
10 MHz to 3 GHz	+18 dBm	+24 dBm (+26 dBm)
3 to 5 GHz	+16 dBm	+19 dBm (+20 dBm)
5 to 6.0 GHz	+16 dBm	+18 dBm (+19 dBm)

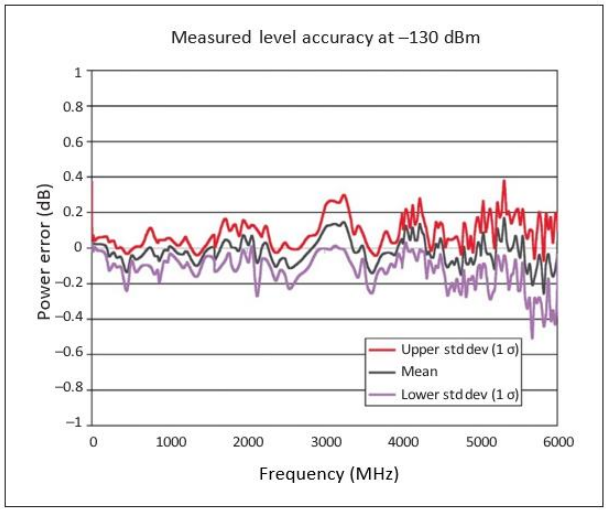
1. Quoted specifications between 20 °C and 30 °C. Maximum output power typically decreases by 0.01 dB/°C for temperatures outside this range.

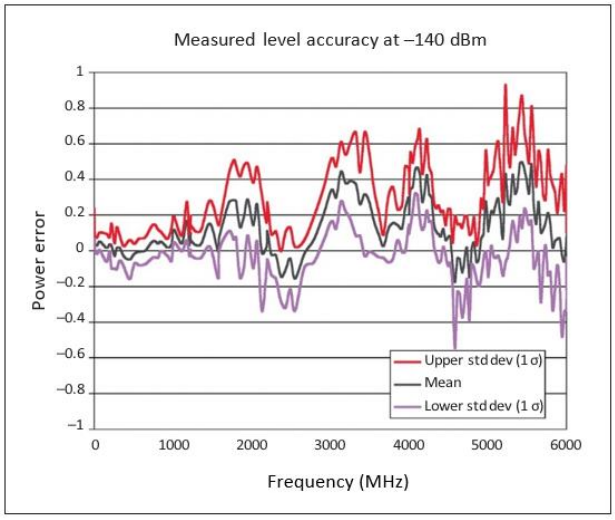


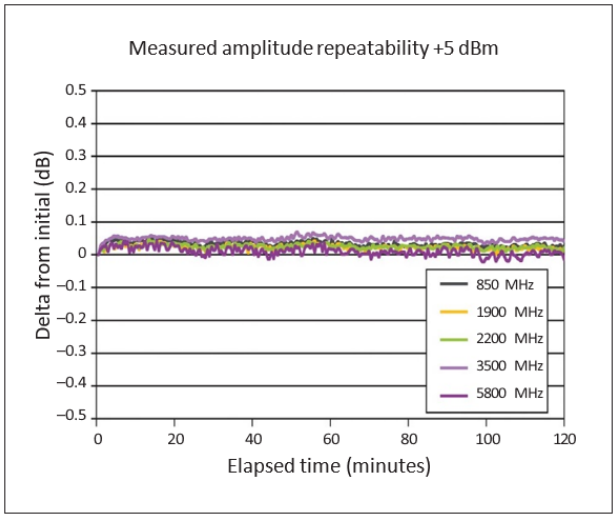
Absolute level accuracy in CW mode [1] (ALC on) ()= typical			
	Standard		Option 1EQ
Range	Max power to –60 dBm	< –60 to –110 dBm	< –110 to –127 dBm
9 to 100 kHz	(± 0.6 dB)	(± 0.9 dB)	
100 kHz to 5 MHz	± 0.8 dB (± 0.3)	± 0.9 dB (± 0.3)	
5 MHz to 3 GHz	± 0.6 dB (± 0.3)	± 0.8 dB (± 0.3)	± 1.5 dB (± 0.5)
3 to 6 GHz	± 0.6 dB (± 0.3)	± 1.1 dB (± 0.3)	± 1.6 dB (± 0.6)
Absolute level accuracy in CW mode (ALC off, power search run, relative to ALC on)			
9 kHz to 6 GHz	± 0.15 dB, typical		
Absolute level accuracy in digital I/Q mode (N5182B only)			
(ALC on, relative to CW, W-CDMA 1 DPCH configuration < +10 dBm)			
5 MHz to 6 GHz	± 0.25 dB, (0.05 dB)		

1. Quoted specifications between 20 °C and 30 °C. For temperatures outside this range, absolute level accuracy degrades by 0.01 dB/°C. Output power may drift up to 0.10 dB < 3 GHz and 0.15 dB > 3 GHz per g/kg change in absolute humidity (nom).

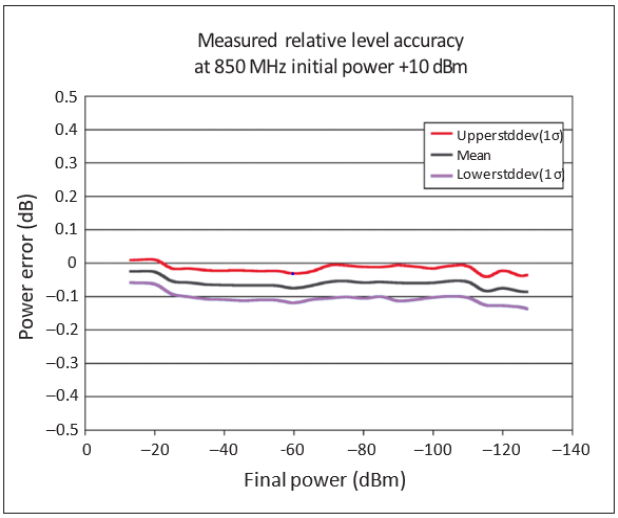




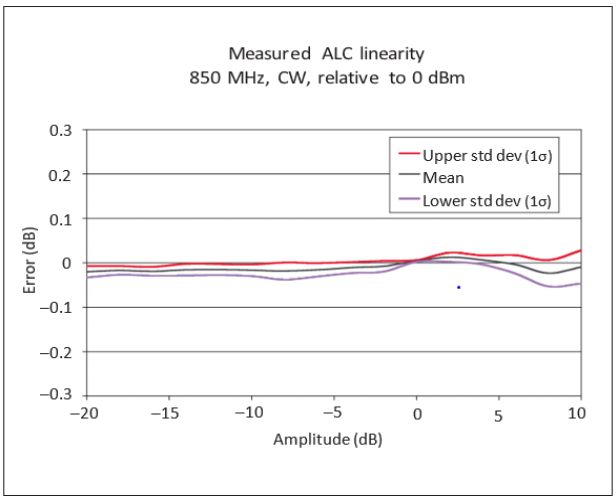


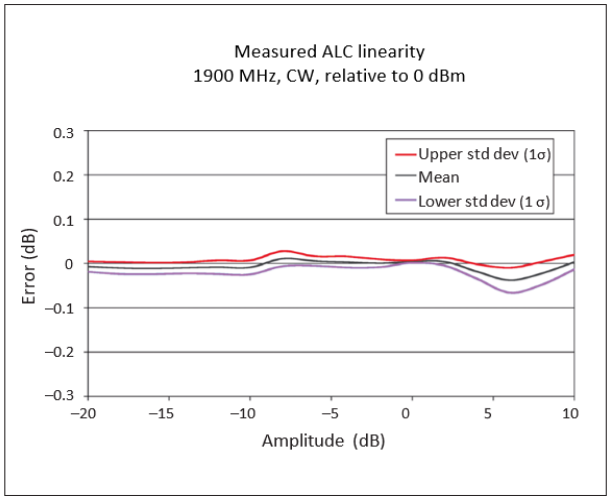


Repeatability measures the ability of the instrument to return a given power setting after a random excursion to any other frequency and power setting. It should not be confused with absolute level accuracy.



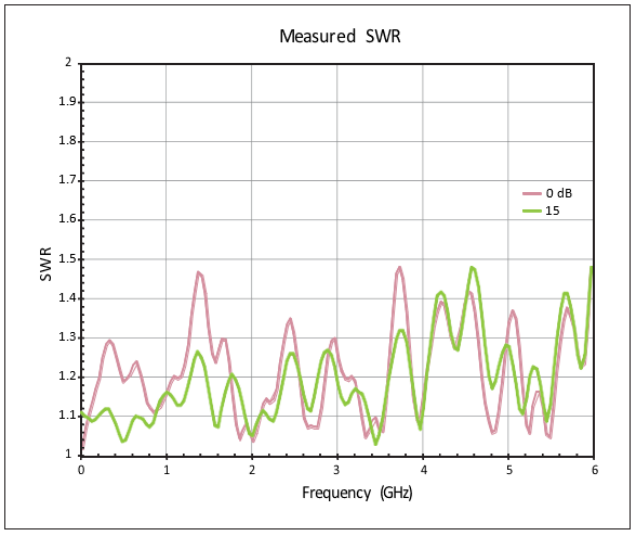
Relative level accuracy measures the accuracy of a step change from any power level to any other power level. This is useful for large changes (such as 5 dB steps).

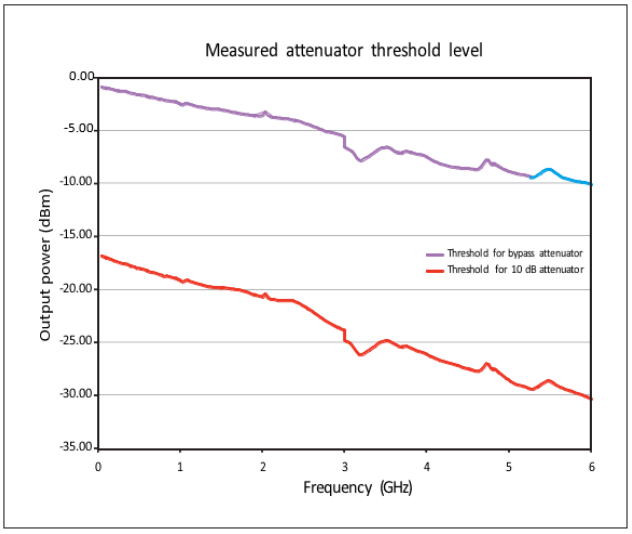




SWR (measured CW mode) [1]			
Frequency	Attenuator state		
	Bypass	0 to 10 dB	15 dB or more
≤ 1.0 GHz	< 1.3:1	< 1.35:1	< 1.2:1
1.0 to 2 GHz	< 1.55:1	< 1:5:1	< 1.3:1
2 to 3 GHz	< 1.8:1	< 1.5:1	< 1.45:1
3 to 4 GHz	< 1.5:1	< 1.6:1	< 1.7:1
4 to 6 GHz	< 1.9:1	< 1.6:1	< 1.6:1

1. SWR < 1.60:1 below 30 kHz.





Maximum reverse power, nominal			
< 1 GHz	50 W		
1 to 2 GHz	25 W		
2 to 6 GHz	20 W		
Max DC voltage	50 VDC		
Trip level	2 W		
Amplitude switching speed [1]	**Standard**	**Option UNZ**	**Option UNZ, typical**
CW mode			
SCPI mode	≤ 5 ms, typical	≤ 750 µs	≤ 650 µs
Power search SCPI mode	< 12 ms, measured		
List/step sweep mode	≤ 5 ms, typical	≤ 500 µs	≤ 300 µs
Digital modulation on (N5182B only)			
SCPI mode	≤ 5 ms, typical	≤ 1.15 ms	≤ 950 µs
Power search SCPI mode	< 12 ms, measured		
List/step sweep mode	≤ 5 ms, typical	≤ 900 µs	≤ 400 µs
Alternate power level control (N5182B only)			
Switching time (via waveform markers)	20 µs within ± 1 dB, measured		
Functional power range	–15 dBm to –144 dBm, measured		
User flatness correction			
Number of points	3201		
Number of tables	Dependent on available free memory in instrument; 10,000 maximum		
Entry modes	USB/LAN direct power meter control, LAN to GPIB and USB to GPIB, remote bus and manual USB/GPIB power meter control		
Sweep modes			
See Frequency Specifications section for more detail			

1. Time from receipt of SCPI command or trigger signal to amplitude settled within 0.2 dB. Switching speed specifications apply when status register updates are off.

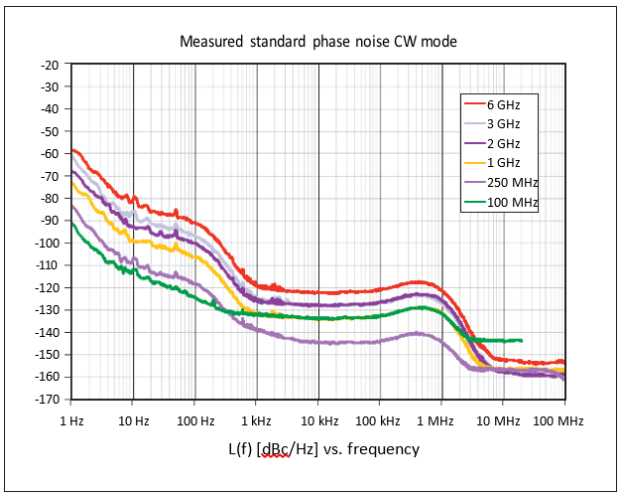
Spectral Purity Specifications

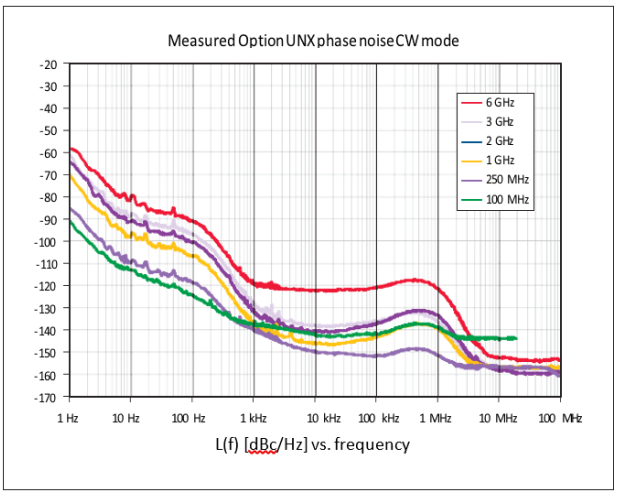
Standard absolute SSB phase noise (dBc/Hz, CW, at 20 kHz offset) () = typical [1]	
5 MHz to < 250 MHz	–129 (–133)
250 MHz	–140 (–143)
500 MHz	–135 (–139)
1 GHz	–131 (–134)
2 GHz	–124 (–127)
3 GHz	–123 (–127)
4 GHz	–118 (–122)
6 GHz	–116 (–121)
Option UNX absolute SSB phase noise (dBc/Hz, CW, at 20 kHz offset) () = typical [1]	
5 MHz to < 250 MHz	–140 (–143)
250 MHz	–144 (–150)
500 MHz	–143 (–150)
1 GHz	–141 (–146)
2 GHz	–135 (–141)
3 GHz	–131 (–137)
4 GHz	–118 (–122)
6 GHz	–117 (–121)

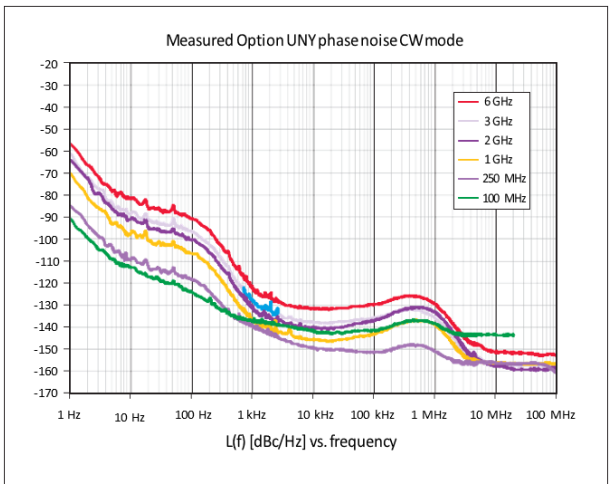
1. From 20 to 30 °C, excludes mechanic vibration, measured @ +10 dBm or maximum specified power, whichever is less.

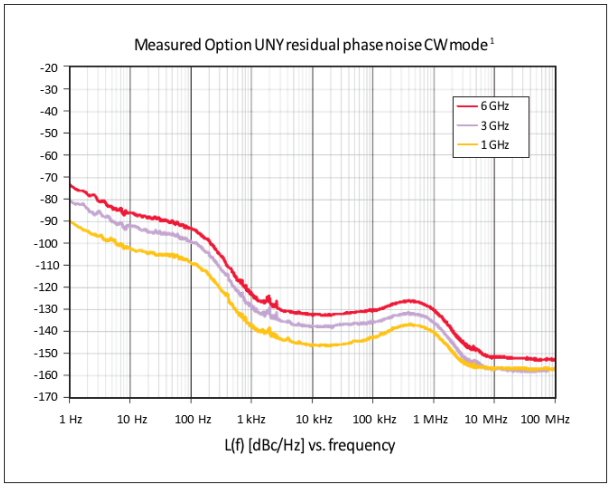
Option UNY absolute SSB phase noise (CW) () = measured [1]						
Frequency	**1 Hz**	**10 Hz**	**100 Hz**	**1 kHz**	**10 kHz**	**100 kHz**
100 MHz	(–91)	(–113)	(–124)	(–137)	(–142)	(–142)
249 MHz	(–85)	–93 (–110)	–103 (–118)	–130 (–137)	–139 (–142)	–138 (–142)
250 MHz	(–85)	–96 (–110)	–104 (–118)	–127 (–139)	–144 (–150)	–147 (–152)
500 MHz	(–74)	–89 (–100)	–98 (–109)	–125 (–139)	–139 (–149)	–145 (–149)
1 GHz	(–70)	–87 (–97)	–93 (–106)	–123 (–136)	–141 (–146)	–140 (–143)
2 GHz	(–65)	–79 (–90)	–85 (–101)	–114 (–131)	–135 (–140)	–134 (–137)
3 GHz	(–61)	–74 (–88)	–81 (–98)	–112 (–128)	–132 (–138)	–131 (–135)
4 GHz	(–61)	–73 (–84)	–79 (–95)	–110 (–124)	–130 (–134)	–127 (–131)
6 GHz	(–57)	–69 (–81)	–76 (–91)	–107 (–121)	–126 (-132)	–125 (–129)

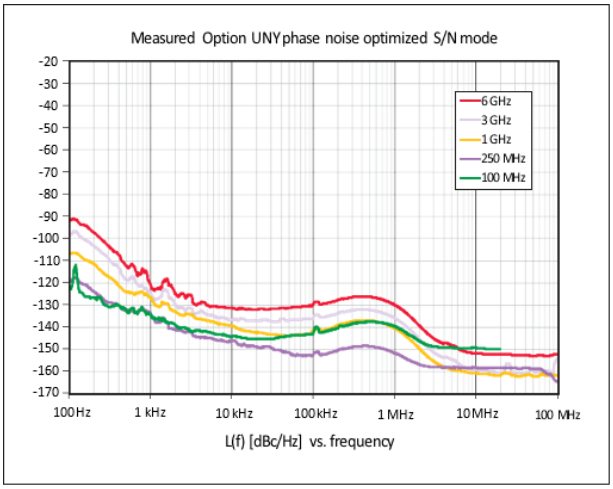
1. From 20 to 30 °C, excludes mechanic vibration, measured @ +10 dBm or maximum specified power, whichever is less.

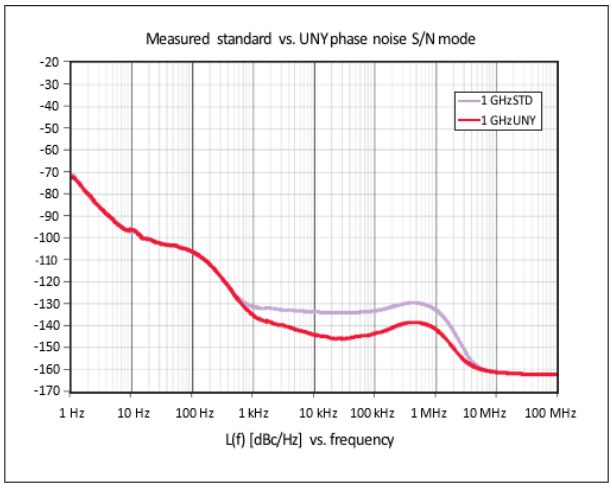


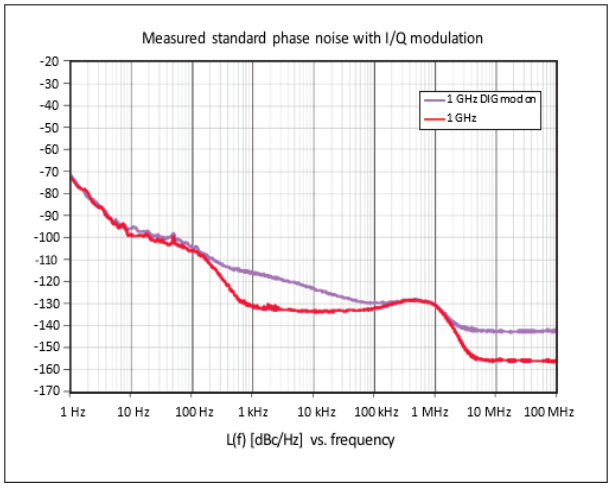


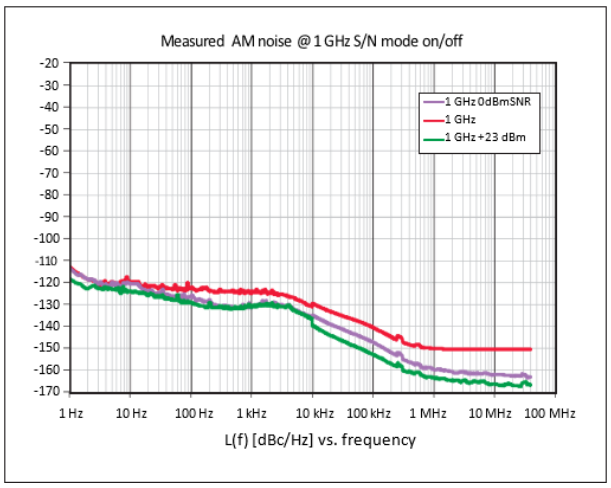












1. Use external 10 MHz input path, between +3 to +7 dBm for maximum performance.

Residual FM (CW mode, 300 Hz to 3 kHz BW, CCITT, rms)		
5 MHz to 6 GHz	< N x 2 Hz (measured) (see N value in frequency band table)	
Residual AM (CW mode, 0.3 to 3 kHz BW, rms, +5 dBm)		
100 kHz to 3 GHz	< 0.01% (measured)	
Harmonics (CW mode)		
Range	**Standard < +4 dBm**	**Option 1EA < +12 dBm**
9 kHz to 3 GHz	< –35 dBc	< –30 dBc
3 to 4 GHz	< –35 dBc, typical	< –35 dBc, typical
4 to 6 GHz	< –53 dBc, typical	< –40 dBc, typical
Nonharmonics (CW mode) [1] () = typical		
Range	**10 KHz offset**	
	Standard (dBc)	**UNX or UNY (dBc)**
9 kHz to < 5 MHz	–65, nominal	–65, nominal
5 to < 250 MHz	–75	–75 (–80)
250 to < 750 MHz	–87	–96 (–100)
750 MHz to < 1.5 GHz	–87	–92 (–96)
1.5 to < 3.0 GHz	–81	–86 (–90)
3 to 6 GHz	–75	–80 (–84)
Subharmonics (CW mode) () = typical		
9 kHz to 1.5 GHz	None	
1.5 to 3 GHz	–77 dBc (–91)	
3 to 6 GHz	–74 dBc (–81)	

1. < 3 GHz fixed 100 MHz spur is specified @ –78 dBc. In signal-to-noise optimization mode 100 MHz spur is < –100 dBc, measured.

Jitter (standard phase noise) [1]				
Carrier frequency	**SONET/SDH data rate**	**rms jitter BW**	**µUI rms, typical**	**Seconds, typical**
155 MHz	155 MB/s	100 Hz to 1.5 MHz	91.8	0.6 ps
622 MHz	622 MB/s	1 KHz to 5 MHz	50.5	81 fs
2.488 GHz	2488 MB/s	5 kHz to 20 MHz	198	80 fs
Jitter (UNX or UNY phase noise) [1]				
Carrier frequency	**SONET/SDH data rate**	**rms jitter BW**	**µUI rms, measured**	**Seconds, measured**
155 MHz	155 MB/s	100 Hz to 1.5 MHz	40	0.25 ps
622 MHz	622 MB/s	1 KHz to 5 MHz	21	33 fs
2.488 GHz	2488 MB/s	5 kHz to 20 MHz	72	29 fs

Phase coherence (Option 012)	
LO input frequency range	250 MHz to 6 GHz, nominal
LO input power range	0 to +12 dBm, nominal
LO output frequency range	250 MHz to 6 GHz, nominal
LO output power range	0 to +12 dBm, nominal

1. Calculated from phase noise performance in CW mode at +10 dBm. For other frequencies, data rates, or bandwidths, please consult your sales representative.

Analog Modulation Specifications

Frequency bands		
Band #	**Frequency range**	**N**
1	9 kHz to < 5 MHz	(digital synthesis)
1	5 to < 250 MHz	1
2	250 to < 375 MHz	0.25
3	375 to < 750 MHz	0.5
4	750 to < 1500 MHz	1
5	1500 to < 3000.001 MHz	2
6	3000.001 to 6000 MHz	4

Frequency modulation (Option UNT) (See N value above)		
Max deviation	N × 4 MHz, nominal [3]	
Resolution	1 Hz, nominal	
Deviation accuracy	< ± 2% + 20 Hz (1 kHz rate, deviation is N x 50 kHz)	
Modulation frequency response @ 100 kHz deviation	1 dB bandwidth 3 dB bandwidth	DC/5 Hz to 3 MHz, nominal DC/1 Hz to 7 MHz, nominal
Carrier frequency accuracy	< ± 0.2% of set deviation + (N × 1 Hz) [1]	
Relative to CW	< ± 0.06% of set deviation + (N × 1 Hz), typical [2]	
Total harmonic distortion	< 0.4% [1 kHz rate, deviation is N x 50 kHz]	
FM using external inputs 1 or 2	Sensitivity	+1 V peak for indicated deviation, nominal
	Input impedance	50 Ω/600 Ω/1 MΩ, nominal
	Paths	FM path 1 and FM path 2 are summed internally for composite modulation

1. Specification valid for temperature changes of less than ± 5 °C since last DCFM calibration.
2. Typical performance immediately after a DCFM calibration.
3. Digital synthesis band FM deviation is 5 MHz.

Phase modulation (Option UNT) (See N value above)			
Maximum deviation	Normal bandwidth	N × 2 radians, nominal	
	High-bandwidth mode	N × 0.2 radians, nominal	
Frequency response	Normal bandwidth (3 dB)	DC to 1 MHz, nominal	
	High-bandwidth mode (3 dB)	DC to 4 MHz, nominal	
Resolution	0.1% of deviation		
Deviation accuracy	< + 0.5% + 0.01 rad, typical [1 kHz rate, normal bandwidth mode]		
Total harmonic distortion	< 0.2%, typical [1 kHz rate, N x 1 radian deviation normal bandwidth mode]		
ΦM using external inputs 1 or 2	Sensitivity	+1 V peak for indicated deviation, nominal	
	Input impedance	50 Ω or 600 Ω or 1 MΩ, nominal	
	Paths	ΦM path 1 and ΦM path 2 are summed internally for composite modulation	
Amplitude modulation (Option UNT) [1]			
AM depth type	Linear or exponential		
Maximum depth	100%		
Depth resolution	0.1% of depth (nom)		
AM depth error @1 kHz rate and < 80% depth	f < 5 MHz	< 1.5% of setting + 1% (typ 0.5% of setting + 1%)	
	5 MHz < f < 2 GHz	< 3% of setting + 1 %	
	2 < f < 3 GHz	< 5% of setting + 1% (typical 3% of setting + 1%)	
Total harmonic distortion @ 1 KHz rate	F < 5 MHz	30% depth	< 0.25%, typical
		80% depth	< 0.5%, typical
	5 MHz < f < 2 GHz (2 to 3 GHz is typical)	30% depth	< 2%
		80% depth	< 2%
Frequency response	30% depth, 3 dB BW	DC/10 Hz to 50 KHz	
Frequency response wideband AM (N5182B only)	Rates ALC off/on:	DC/800 Hz to 80 MHz, nominal	

1. AM specifications apply 6 dB below maximum specified power from 20 to 30 °C.

AM inputs using external inputs 1 or 2	Sensitivity	± 1 V peak for indicated depth (Over-range can be 200% or 2.2 V peak)
	Input impedance	50 Ω or 600 Ω or 1M Ω, Damage level: ± 5 V max
	Paths	AM path 1 and AM path 2 are summed internally for composite modulation
Wideband AM inputs (N5182B only)	Sensitivity	1 V peak-to-peak sine wave signal with 0.5 V DC offset required input for 100% AM
	Input impedance	50 Ω, nominal (I input)

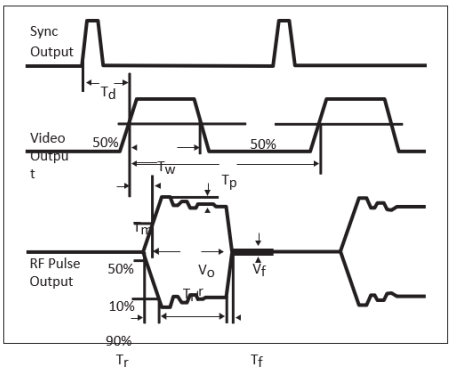
Simultaneous and composite modulation [2]	
Simultaneous modulation	All modulation types (I/O, FM, AM, ΦM, and pulse modulation) may be simultaneously enabled except: FM and phase modulation cannot be combined and two modulation types cannot be simultaneously generated using the same modulation source; for example, the baseband I/Q generator, AM, and FM can run concurrently and all will modulate the output RF (this is useful for simulating signal impairments)
Composite modulation	AM, FM, and ΦM each consist of two modulation paths which are summed internally for composite modulation; modulation can be any combination of internal or external sources

	AM	FM	Phase	Pulse	Internal I/Q [1]	External I/Q [1]
AM	+	+	+	+	+	+
FM	+	+	–	+	+	+
Phase	+	–	+	+	+	+
Pulse	+	+	+	–	+	+
Internal I/Q (1)	+	+	+	+	–	+
External I/Q (1)	+	+	+	+	+	–

+ = compatible, - = incompatible, * = Internal + External

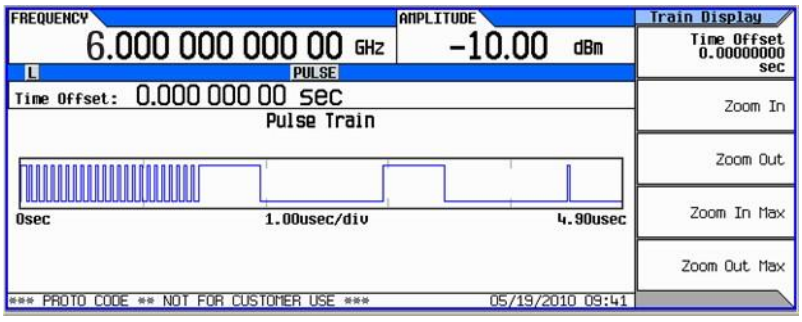
1. AM specifications apply 6 dB below maximum specified power from 20 to 30 °C.
2. I/Q modulation available on N5182B.

External modulation inputs	
(Option UNT required for FM, AM, and phase modulation inputs; Option UNW required for pulse modulation inputs)	
EXT1	AM, FM, PM
EXT2	AM, FM, PM
PULSE	Pulse (50 Ω only)
I	Wideband AM (50 Ω only, N5182B only requires Q to be biased with 1.0 V)
Input impedance	50 Ω, 1 MΩ, 600 Ω, DC and AC coupled
Standard internal analog modulation source	
(Single sine wave generator for use with AM, FM, phase modulation requires Option UNT or 303)	
Waveform	Sine, square, triangle, positive ramp, negative ramp
Rate range	0.1 Hz to 2 MHz (tunable to 3 MHz)
Resolution	0.1 Hz
Frequency accuracy	Same as RF reference source, nominal
LF audio output	0 to 5 V peak into 50 Ω, –5V to 5 V offset, nominal
Multifunction generator (Option 303)	
The multifunction generator option (Option 303) consists of seven waveform generators that can be set independently with up to five simultaneously using the composite modulation features in AM, FM/PM, and LF out	
Waveform	
Function generator 1	Sine, triangle, square, positive ramp, negative ramp, pulse
Function generator 2	Sine, triangle, square, positive ramp, negative ramp, pulse
Dual function generator	Sine, triangle, square, positive ramp, negative ramp, phase offset, and amplitude ratio for Tone 2 relative to Tone 1
Swept function generator	Sine, triangle, square, positive ramp, negative ramp Trigger: free run, trigger key, bus, external, internal, timer trigger
Noise generator 1	Uniform, Gaussian
Noise generator 2	Uniform, Gaussian
DC	Only for LF output –5 V to +5 V, nominal

Frequency parameters	
Sine wave	0.1 Hz to 10 MHz
Triangle, square, ramp, pulse	0.1 Hz to 1 MHz, nominal
Noise bandwidth	10 MHz, nominal
Resolution	0.1 Hz
Frequency accuracy	Same as RF reference source, nominal
Narrow pulse modulation (Option UNW) [1] () = typical	
On/off ratio	(> 80 dB)
Rise/fall times (T_r, Tf)	< 10 ns; (7 ns)
Minimum pulse width ALC on/off	> 2 µs/> 20 ns
Repetition frequency ALC on/off	10 Hz to 500 kHz/DC to 10 MHz
Level accuracy (relative to CW) ALC on/off [2]	< ± 1.0 (± 0.5) dB/(< ± 0.5) dB
Width compression (RF width relative to video out)	(< 5 ns)
Video feed-through [3] ≤ 3 GHz/> 3 GHz	(< 50 mV/< 5 mV)
External video delay (ext input to video)	30 ns, nominal
RF delay (video to RF output)	20 ns, nominal
Pulse overshoot	(< 15%)
Input level	+1 Vpeak = RF on into 50 Ω, nominal
T_d video delay (variable) T_w video pulse width (variable) T_p pulse period (variable) T_m RF delay T_{rf} RF pulse width T_f RF pulse fall time T_r RF pulse rise time V_{or} pulse overshoot V_f Video feedthrough	

1. Pulse specifications apply to frequencies > 100 MHz and power set to > -3 dBm. Operable down to 9 kHz.
2. With power search on.
3. Video feed through applies to power levels < +10 dBm.

Internal pulse generator (included with Option UNW)		
Modes	Free-run, square, triggered, adjustable doublet, trigger doublet, gated, and external pulse	
Square wave rate	0.1 Hz to 10 MHz, 0.1 Hz resolution, nominal	
Pulse period	30 ns to 42 seconds, nominal	
Pulse width	20 ns to pulse period –10 ns, nominal	
Resolution	10 ns	
Adjustable trigger delay	(– pulse period + 10 ns) to (pulse width –10 ns)	
Settable delay	Free run	–3.99 to 3.97 µs
	Triggered	0 to 40 s
Resolution (delay, width, period)	10 ns, nominal	
Pulse doublets	1st pulse delay	(Relative to sync out) 0 to 42 s – pulse width – 10 ns
	1st pulse width	20 ns to 42 s – delay – 10 ns
	2nd pulse delay	0 to 42 s – (Delay 1 + Width 2) – 10 ns
	2nd pulse width	20 ns to 42 s – (Delay 1 + Delay 2) – 10 ns
Pulse train generator Option N5180320B (requires Option UNW)		
Number of pulse patterns	2047	
On/off time range	20 ns to 42 sec	



Avionics (Option N5180302B)		
VOR		
Bearing accuracy		± 0.1°
Frequency accuracy		Same as RF reference source, nominal
AM accuracy	30% depth	± 5% of setting
AM distortion		2%
FM accuracy	480 Hz deviation	± 1.7 Hz
ILS: localizer and glide slope		
AM accuracy	40% depth	± 5% of setting
AM distortion		2%
Difference in depth of modulation (DDM) resolution	Localizer	0.0002
	Glide slope	0.0004
Difference in depth of modulation (DDM) accuracy	Localizer	± 0.0004 ± 5% of DDM [1]
	Glide slope	± 0.0008 ± 5% of DDM [1]
Marker beacon		
Marker tone AM accuracy	95% depth	± 5% of setting + 1%
Marker tone AM distortion	95% depth	5%

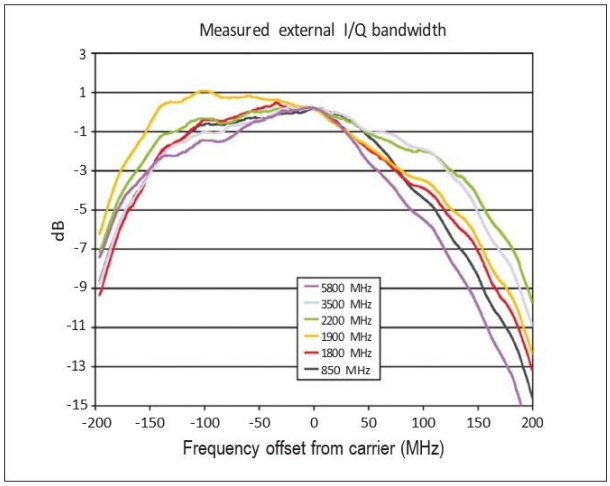
1. DDM must not be equal to 0.

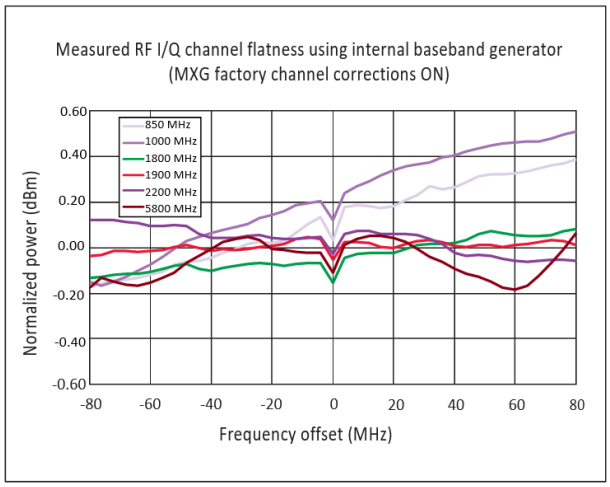
Vector Modulation Specifications

N5182B Only

I/Q modulator external inputs		
Bandwidth	Baseband (I or Q)	Up to 100 MHz baseband, nominal
	RF (I+Q)	Up to 200 MHz RF
I or Q offset	± 100 mV (200 uV resolution)	
I/Q gain balance	± 4 dB (0.001 dB resolution)	
I/Q attenuation	0 to 50 dB (0.01 dB resolution)	
Quadrature angle adjustment	± 200 units (0.1 units resolution)	
Full scale input drive (I+Q)	0.5 V into 50 Ω, nominal	
Internal I/Q baseband generator adjustments [1, 2] (Options 656 and 657)		
I/Q offset	± 20% (0.025% resolution)	
I/Q gain	± 1 dB (0.001 dB resolution)	
Quadrature angle adjustment	± 10 ° (0.01 degrees resolution)	
I/Q phase	± 360.00 ° (0.01 degrees resolution)	
I/Q skew	± 800.00 ns (1 picosecond resolution)	
I/Q delay	± 250.00 ns (1 picosecond resolution)	
External I/Q outputs [1]		
Impedance	50 Ω, nominal per output	
	100 Ω, nominal differential output	
Type	Single-ended or differential (Option 1EL)	
Maximum voltage per output	1 V peak-to-peak or 0.5 V peak	
Bandwidth (I, Q)	Baseband (I or Q)	80 MHz, nominal (Option 656 and 657)
	RF (I+Q)	160 MHz, nominal (Option 656 and 657)
Amplitude flatness	± 0.2 dB measured with channel corrections optimized for I/Q output	
Phase flatness	± 2.5 degrees measured with channel corrections optimized for I/Q output	
Common mode I/Q offset	± 1.5 V into 50 Ω (200 µV resolution)	
Differential mode I or Q offset	± 50 mV into 50 Ω (200 µV resolution)	

1. I/Q adjustments represent user interface nominal parameter ranges and not specifications.
2. Internal I/Q adjustments apply to RF out and I/Q outputs simultaneously.





Internal real-time complex digital I/Q filters (included with Option 656)	
Factory channel correction (256 taps)	
Corrects the linear phase and amplitude response of the baseband I/Q and RF outputs of the signal generator using factory calibration arrays. (default mode is off)	
RF amplitude flatness (160 MHz)	± 0.2 dB measured
RF phase flatness (160 MHz)	± 2 degrees measured
User channel correction (256 taps)	
Automated routine uses power sensor to correct for linear phase and amplitude response of DUT (equalizer). See Users Guide for more details.	
Recommended max amplitude error for correction	± 15 dB
Recommended max phase error for correction	± 25 degrees
Equalization filter (256 taps)	
User can download and apply inverse or custom phase and amplitude response coefficients from tools such as MATLAB, 89600 VSA or SystemVue to correct for linear errors of DUT/system. See Users Guide for more details.	

Baseband generator (Options 656 and 657)		
Channels	2 [I and Q]	
Resolution	16 bits [1/65,536]	
Sample rate	Option 656	100 Sa/s to 100 MSa/s
	Option 656 and 657	100 Sa/s to 200 MSa/s
Maximum number of waveform files in cache	1024	
RF (I+Q) bandwidth	Option 656 Option 656 and 657	80 MHz, nominal 160 MHz, nominal
Interpolated DAC rate	800 MHz (waveforms only need OSR = 1.25)	
Frequency offset range	± 80 MHz	
Digital sweep modes	In list sweep mode each point in the list can have independent waveforms (N5182B) along with user definable frequencies and amplitudes; see the Amplitude and Frequency Specifications sections for more detail.	
Waveform switching speed [1]	SCPI mode	≤ 5 ms, measured (standard)
		≤ 1.2 ms, measured (Option UNZ)
	List/step sweep mode	≤ 5 ms, measured (standard)
		≤ 900 µs, measured (Option UNZ)
Waveform transfer rates (measured, no markers, unencrypted)	FTP LAN to internal SSD	10.7 MB/sec or 2.67 Msa/sec
	Internal SSD to FTP LAN	7.7 MB/sec 1.92 Msa/sec
	FTP LAN to BBG	8.2 MB/sec or 2.05 Msa/sec
	FTP LAN to BBG encrypted	4 MB/sec or 1 Msa/sec
	USB to BBG	19 MB/sec or 4.75 Msa/sec
	BBG to USB	1.2 MB/sec or 300 Ksa/sec
	Internal SSD to BBG	48 MB/sec or 12 Msa/sec
	BBG to internal SSD	1.2 MB/sec or 300 Ksa/sec
	SD card to BBG (Option 006)	2.7 MB/sec or 678 Ksa/sec
	BBG to SD card (Option 006)	845 KB/sec or 211 Ksa/sec

1, SCPI mode switching speed applies when waveforms are pre-loaded in list sweep and sample rate ≥ 10 MSa/s.

Arbitrary waveform memory	Maximum playback capacity		32 Msa (standard)
			512 Msa (Option 022)
			1024 Msa (Option 023)
	Maximum storage capacity including markers		3 GBytes/800 Msa (standard)
			30 GBytes/7.5 Gsa (Option 009)
			8 GBytes / 2 Gsa (Option 006)
Waveform segments	Segment length		60 samples to 32 Msa (standard)
			60 samples to 512 Msa (Option 022)
			60 samples to 1024 Msa (Option 023)
	Minimum memory al- location per segment		256 samples
	Maximum number of segments		8192
	Label		Maximum number of waveform files
	Value		1024
Waveform sequences	Maximum number of sequences		2000 depending on non-volatile memory usage
	Maximum number of segments/sequence		32,000 (standard)
			4 million (Option 022 or 023)
	Maximum number of repetitions		65,535
Triggers	Types		Continuous, single, gated, segment advance
	Source		Trigger key, external, bus (GPIB, LAN, USB)
	Modes	Continuous	Free run, trigger and run, reset and run
		Single	No retrigger, buffered trigger, restart on trigger
		Gated	Negative polarity or positive polarity
		Segment advance	Single or continuous
	External coarse delay time		5 ns to 40 s
	External coarse delay resolution		5 ns
	Trigger latency (Single trigger only)		356 ns + 1 sample clock period, nominal
	Trigger accuracy (Single trigger only)		± 2.5 ns, nominal
	Single trigger - restart on trigger mode will initiate a FIFO clear. Therefore, the latency includes re-filling the buffer. The latency is 8 µs + (1406 x sample period) ± 1 sample clock period, nominal		

Multi-baseband generator synchronization mode (multiple sources)	Fan out	1 primary and up to 15 secondary
	Trigger repeatability	< 1 ns, nominal
	Trigger accuracy	Same as normal mode
	Trigger latency	Same as normal mode
	Fine trigger delay range	See Internal I/Q Baseband section
	Fine trigger delay resolution	See Internal I/Q Baseband section
	I/Q phase adjustment range	See Internal I/Q Baseband section
Markers	Markers are defined in a segment during the waveform generation process, or from the front panel; a marker can also be routed to the RF blanking, ALC hold functions, and alternate amplitude; see Users Guide for more information	
	Marker polarity	Negative, positive
	Number of markers	4
	RF blanking/burst on/off ratio	80 dB
	Alternate amplitude control switching speed	See amplitude section
Real-time modulation FIR filter:	Filter types: Nyquist, root-Nyquist, WCDMA, EDGE, Gaussian, rectangular, APCO 25 C4FM, IS-95, User FIR (Applies real-time FIR filtering when playing waveforms with OSR=1. Helps reduce waveform size for long simulation times. Option 660 not required.)	
Real-time baseband generator (Option 660)		
Real-time baseband generator required for real-time Signal Studio applications [1]	Cellular real-time applications	LTE-FDD, LTE-TDD, HSPA+/W-CDMA, GSM/EDGE, cdma2000®
	Real-time navigation	GPS, GLONASS, Galileo
	Real-time video applications	DVB-T/T2/H/S/S2/C/J.83 Annex A/C, ISDB-T/
	Note: Option 660 is not required for real-time custom modulation (Option N5180431B)	
	Memory: Shares memory with Options 656 and 657	
	Triggering: Same as Options 656 and 657	
	Markers: 3 markers available, all other features are same as Options 656 and 657	

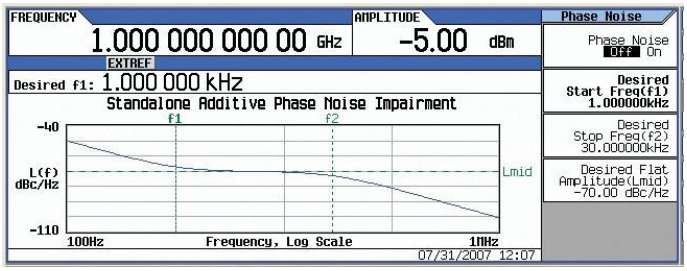
1. See www.keysight.com/find/signalstudio for more information.

Digital baseband inputs/outputs (Option 003/004)		
Options 003 and 004 activate the rear panel digital I/Q bus and enable connectivity to the N5102A digital signal interface module. In output mode (003), you can deliver realistic complex-modulated signals such as LTE, GPS, WLAN, custom pulses and many others directly to your digital devices and subsystems. In the input mode (004), the interface module ports your digital input to the signal generator's baseband system, providing a quick and easy way of upconverting to calibrated analog I/Q, IF, or RF frequencies. In both operating modes, the interface module adapts to your device with the logic type, data format, clock features, and signaling you require.		
Data (requires N5102A)		
Digital data format	User-selectable: 2's complement or binary offset, I/Q (I, I-bar, Q, Q-bar) or digital IF output (real, imaginary)	
Data port	Dual 16-bit data buses support parallel, parallel I/Q interleaved, parallel Q/I interleaved, or serial port configuration	
N5102A connectors (breakout boards)	144-pin Tyco Z-Dok+ connects to break-out boards (included with N5102A) that interface with the following connector types: 68-pin SCSI, 38-pin dual AMP Mictor, 100-pin dual Samtec, 20-pin dual 0.1 inch headers, 40-pin dual 0.1 inch headers	
Logic types	Single-ended: LVTTL, 1.5V CMOS, 1.8V CMOS, 2.5V CMOS, 3.3.V CMOS	
	Differential: LVDS	
Data output resampling	MXG baseband output is resampled to the arbitrary clock rate set by the user via real-time curve-fit calculations.	
Clock (requires N5102A)		
Clock input	User selectable: internal clock, device under test clock, or external clock (via SMA or breakout board)	
	N5102A SMA Ext Clock In connector: 50 Ω, 0 dBm nominal, 1 to 400 MHz	
Clock output	User selectable: via breakout board or SMA Clock Out connector	
	N5102A SMA Clock Out connector: 2 Vpp into load > 5K Ω from 1 to 100 kHz, 400 mVpp into 50 Ω load from 100 kHz to 400 MHz	
Sample rate (limited by MXG sample rate)	User-selectable in parallel mode up to a maximum 200 MHz, but limited by other user settings (see N5102A users guide for more details).	
	User-selectable in serial mode, the maximum rate is 400 MHz/word size.	
Bit rate (limited by MXG sample rate)	Parallel Up to 200 MHz x word size (1.6 Gbps LVDS, CMOS and LVTTL) per parallel bus, 2 parallel buses available	
	Serial Up to 400 MHz per serial line (400 Mbps LVDS) or 150 MHz per serial line (150 Mbps (CMOS/LVTTL) 32 lines available)	
Clocks per sample	In parallel output mode, the data sample can be held for 1, 2 or 4 clock cycles	
Clock to data skew	Coarse adjustment in 90° steps from 0 to 270°; fine-adjustment in increments of 100 ps up to 5 ns	
Clock polarity	Clock signals may be inverted	
Frequency reference input	1 to 100 MHz BNC, 50 Ω, 3 dBm ± 6 dB	
Power supply (included on N5102A)	Output: 5V, 4A DC	
AWGN (Option N5180403B)		
Type	Real-time, continuously calculated, and played using DSP	
Modes of operation	Standalone or digitally added to signal played by arbitrary waveform or real-time baseband generator	
Bandwidth	With Option 656	1 Hz to 80 MHz
	With Option 656 and 657	1 Hz to 160 MHz
Crest factor	15 dB	

Randomness	90 bit pseudo-random generation, repetition period 313 x 10^9 years		
Carrier-to-noise ratio	± 100 dB when added to signal		
Carrier-to-noise ratio formats	C/N, Eb/No		
Carrier-to-noise ratio error	Magnitude error ≤ 0.2 dB at baseband I/Q outputs		
Custom modulation Arb Mode (Option N5180431B)			
Modulation	PSK		BPSK, QPSK, OQPSK, π/4DQPSK, gray coded and unbalanced QPSK, 8PSK, 16PSK, D8PSK, IS95 QPSK, IS95 OQPSK, EDGE, HDQPSK
	QAM		4, 16, 32, 64, 128, 256, 1024 (and 89600 VSA mappings)
	FSK		Selectable: 2, 4, 8, 16, C4FM, HCPM
	MSK		0 to 100°
	ASK		0 to 100%
	DVB-S2 APSK		16APSK 2/3, 16APSK 3/4 16APSK 4/5, 16APSK 5/6, 16APSK 8/9, 16APSK 9/10, 32APSK 3/4 32APSK 4/5, 32APSK 5/6, 32APSK 8/9, 32APSK 9/10
Multicarrier	Number of carriers		Up to 100 (limited by a max bandwidth of 160 MHz depending on symbol rate and modulation type)
	Frequency offset (per carrier)		Up to –80 to +80 MHz
	Power offset (per carrier)		0 dB to –40 dB
Symbol rate	50 sps to 100 Msps		
Filter types	Nyquist, root-Nyquist, Gaussian, rectangular, APCO 25 C4FM, user		IS-95 w/EQ, IS-95 Mod, IS-95 Mod w/EQ, HDQPSK, APCO25 HCPM, SOQPSK-TG
Quick setup modes	PCO25w/C4FM, APCO25w/CQPSK, *Bluetooth®*, CDPD, DECT, EDGE, GSM, NADC, PDC, PHS, PWT, TETRA		16APSK 2/3, 16APSK 3/4 16APSK 4/5, 16APSK 5/6, 16APSK 8/9, 16APSK 9/10, 32APSK 3/4, 32APSK 4/5,32APSK 5/6, 32APSK 8/9, 32APSK 9/10
Custom modulation real-time mode (Option N5180431B) (Does not require Option 660)			
Modulation	PSK		BPSK, QPSK, OQPSK, π/4DQPSK, gray coded and unbalanced QPSK, 8PSK, 16PSK, D8PSK, SOQPSK
	QAM		4, 16, 32, 64, 128, 256, 1024 (and 89600 VSA mappings)
	FSK	Selectable	2,4,8,16 level symmetric, C4FM
		User-defined	Custom map of up to 16 deviation levels
		Max deviation	20 MHz
	MSK	0 to 100°	
	ASK	0 to 100%	
	Custom I/Q	Custom map of 1024 unique values	
Frequency offset	Up to –80 MHz to +80 MHz		
Symbol rate	Internal generated data		1 sps up to 100 Msps and max of 10 bits per symbol (Option 656 + 657)
	External serial data		1 sps to [(50 Mbits/sec)/(#bits/symbol)]

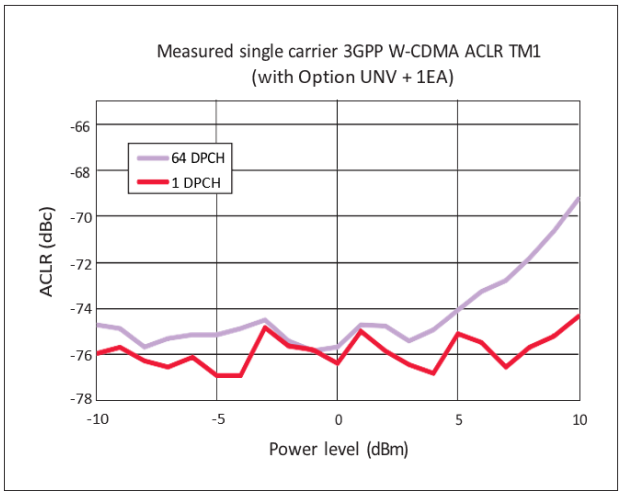
Filter types	Selectable		Nyquist, root-Nyquist, Gaussian, rectangular, APCO 25 (phase 1 and 2 UL and DL), IS-95, WCDMA, EDGE (wide and HSR)
	Custom FIR		16-bit resolution, up to 64 symbols long, automatically resampled to 1024 coefficients (max) 32 to 64 symbol filter: symbol rate ≤ 12.5 MHz 16 to 32 symbol filter: symbol rate ≤ 25 MHz Internal filters switch to 16 tap when symbol rate is between 25 and 100 MHz
Quick setup modes	APCO 25 with (C4FM, CQPSK, HCPM, HDQPSK), TETRA, *Bluetooth*, CDPD, DECT, EDGE, GSM, NADC, PDC, PHS, PWT, WorldSpace, Iridium, ICO, CT2, TFTS, SOQPSK		
Trigger delay	Range		0 to 1,048,575 bits
	Resolution		1 bit
Data types	Internally generated	Pseudo-random patterns	PN9, PN11, PN15, PN20, PN23
		Repeating sequence	Any 4-bit sequence
	Direct-pattern RAM [PRAM] max size Note: Used for custom TDMA/non-standard framing		32 Mb (standard)
			512 Mb (Option 022)
			1024 Mb (Option 023)
	User file		32 MB (standard)
			512 MB (Option 022)
			1024 MB (Option 023)
	Externally streamed data (via AUX I/O)	Type	Serial data
		Inputs/outputs	Data, symbol sync, bit clock
Internal burst shape (varies with bit rate)	Rise/fall time range		Up to 30 bits
	Rise/fall delay range		–15 to +15 bits

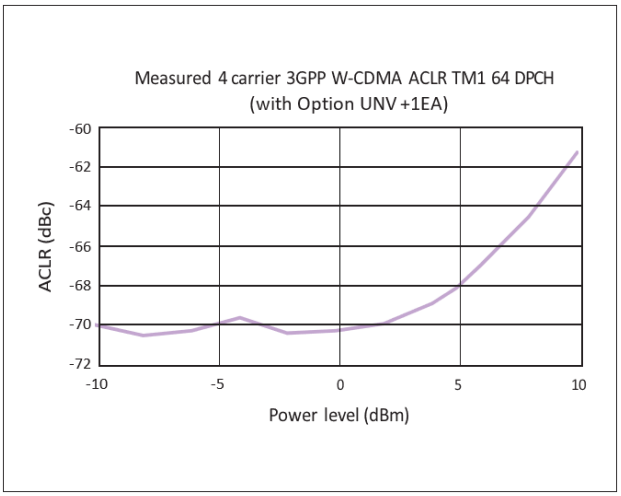
Multitone and two-tone (Option N5180430B)		
Number of tones	2 to 512, with selectable on/off state per tone	
Frequency spacing	100 Hz to 160 MHz (Option 656 and 657)	
Phase (per tone)	Fixed or random	
Real-time phase noise impairments (Option N5180432B)		
Close-in phase noise characteristics	–20 dB per decade	
Far-out phase noise characteristics	–20 dB per decade	
Mid-frequency characteristics	Start frequency (f1)	Offset settable from 0 to 77 MHz
	Stop frequency (f2)	Offset settable from 0 to 77 MHz
Phase noise amplitude level (L(f))	User selected; max degradation dependent on f2	



3GPP W-CDMA distortion performance [1, 2]								
			Standard		Option UNV		Option UNV with Option 1EA	
Power level			≤ 2 dBm [2]		≤ 2 dBm [2]		≤ 5 dBm [2]	
Offset	Configuration	Frequency	Spec	Typ	Spec	Typ	Spec	Typ
Adjacent (5 MHz)	1 DPCH, 1 carrier	1800 to 2200 MHz	– 69 dBc	–73 dBc	–71 dBc	–75 dBc	–71 dBc	–75 dBc
Alternate (10 MHz)			–70 dBc	–75 dBc	–72 dBc	–77 dBc	–71 dBc	–77 dBc
Adjacent (5 MHz)	Test model 1 with 64 DPCH, 1 carrier	1800 to 2200 MHz	–68 dBc	–70 dBc	–71 dBc	–73 dBc	–71 dBc	–72 dBc
Alternate (10 MHz)				–73 dBc	–72 dBc	–76 dBc	–71 dBc	–76 dBc
Adjacent (5 MHz)	Test model 1 with 64 DPCH, 4 carrier	1800 to 2200 MHz	–63 dBc	–65 dBc	–65 dBc	–67 dBc	–64 dBc	–66 dBc
Alternate (10 MHz)			–64 dBc	–66 dBc	–66 dBc	–68 dBc	–66 dBc	–68 dBc

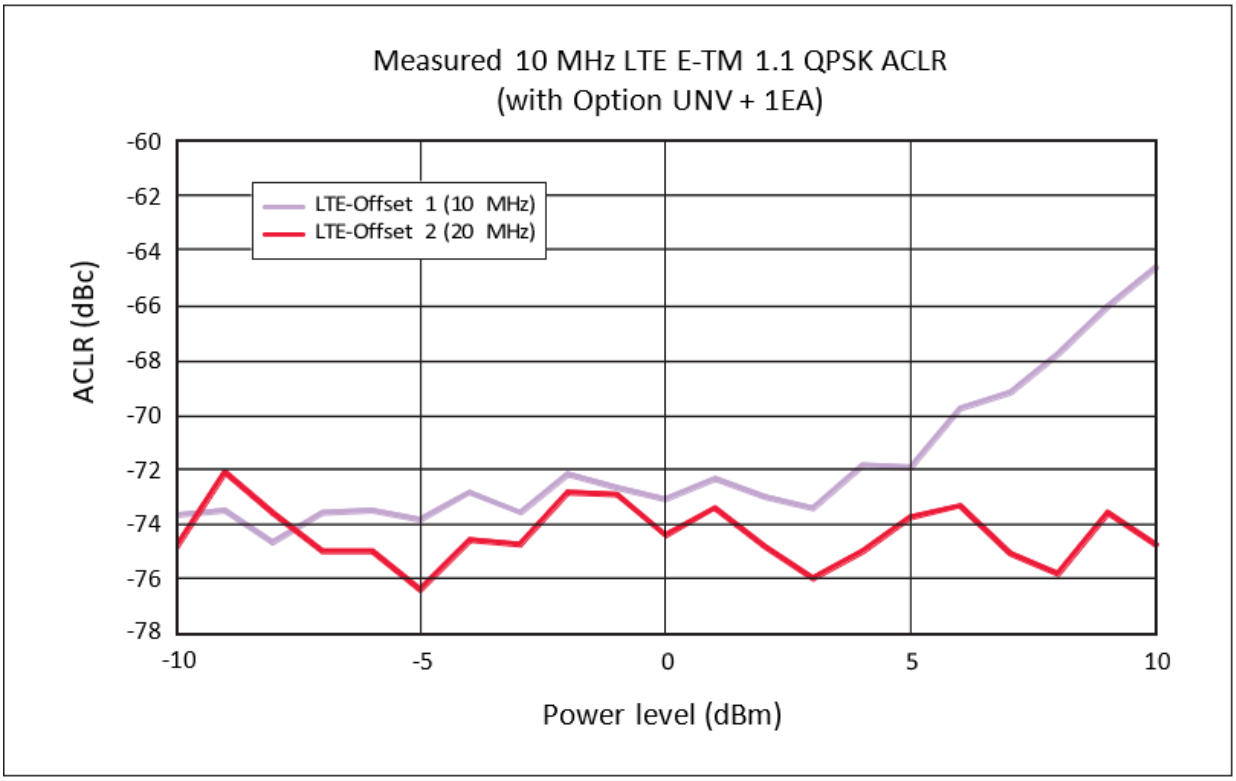
1. ACPR specifications apply when the instrument is maintained within 20 to 30 °C.
2. This is rms power. Convert from rms to peak envelope power (PEP) with the following equation: PEP = rms power + crest factor (for example, 3GPP test model 1 with 64 DPCH has a crest factor 11.5 dB, therefore at +5 dBm rms, the PEP = 5 dBm + 11.5dB = +16.5 dBm PEP).





3GPP LTE-FDD distortion performance [1]								
			Standard		Option UNV		Option UNV with Option 1EA	
Power level			≤ 2 dBm [2]		≤ 2 dBm [2]		≤ 5 dBm [2]	
Offset	Configuration	Frequency	Spec	Typ	Spec	Typ	Spec	Typ
Adjacent (10 MHz) [3]	10 MHz E-TM 1.1 QPSK	1800 to 2200 MHz	–64 dBc	–66 dBc	–67 dBc	–69 dBc	–64 dBc	–67 dBc
Alternate (20 MHz) [3]			–66 dBc	–68 dBc	–69 dBc	–71 dBc	–69 dBc	–71 dBc

1. ACPR specifications apply when the instrument is maintained within 20 to 30 °C.
2. This is rms power. Convert from rms to peak envelope power with the following equation: PEP = rms power + crest factor (for example, 3GPP test model 1 with 64 DPCH has a crest factor 11.5 dB, therefore at +5 dBm rms, the PEP = 5 dBm + 11.5 dB = +16.5 dBm PEP).
3. ACPR measurement configuration: reference channel integration BW: 9.015 MHz, offset channel integration bandwidth: 9.015 MHz.



GSM/EDGE output RF spectrum (ORFS)						
			GSM		EDGE	
Power level			< +7 dBm		< +7 dBm	
Offset	Configuration	Frequency [1]	Standard, typical	Option UNV, typical	Standard, typical	Option UNV, typical
200 kHz	1 normal timeslot, bursted	800 to 900 MHz 1800 to 1900 MHz	–34 dBc	–36 dBc	–37 dBc	–38 dBc
400 kHz			–69 dBc	–70 dBc	–69 dBc	–70 dBc
600 kHz			–81 dBc	–82 dBc	–80 dBc	–81 dBc
800 kHz			–82 dBc	–83 dBc	–82 dBc	–83 dBc
1200 kHz			–84 dBc	–85 dBc	–83 dBc	–84 dBc

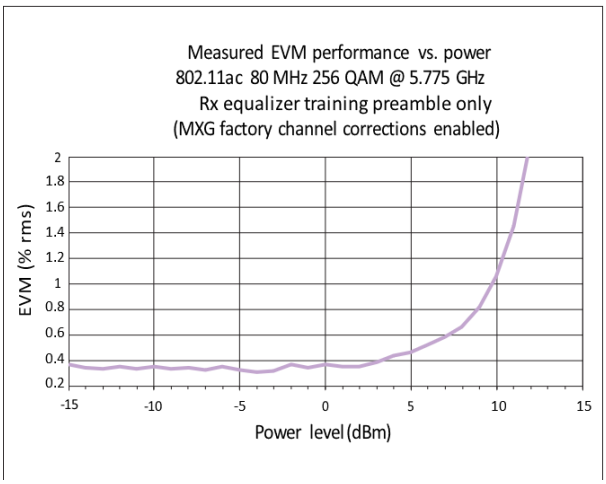
3GPP2 cdma2000 distortion performance, typical					
		Standard	Option UNV	Option UNV + 1EA	
Power level [2]		≤ 2dBm	≤ 2 dBm	≤ 5 dBm	
Offset	Configuration	Frequency (1)	Typical	Typical	Typical
885 kHz to 1.98 MHz	9 channel forward link	800 to 900 MHz	–78 dBc	–79 dBc	–77 dBc
1.98 to 4.0 MHz			–86 dBc	–87 dBc	–87 dBc
> 4.0 to 10 MHz			–91dBc	–93 dBc	–93 dBc

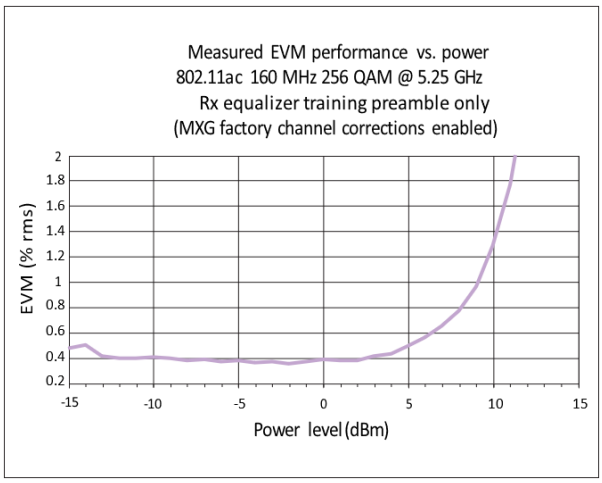
802.16e Mobile WiMAX™ distortion performance, measured					
Power	Offset [3]	Configuration [4]	Frequency	Standard, measured	UNV, measured
< -7 dBm	10 MHz	QPSK	2.5 and 3.5 GHz	–65 dBc	–68 dBc
Up to +5 dBm	10 MHz	QPSK	3.5 GHz	–62 dBc	–65 dBc

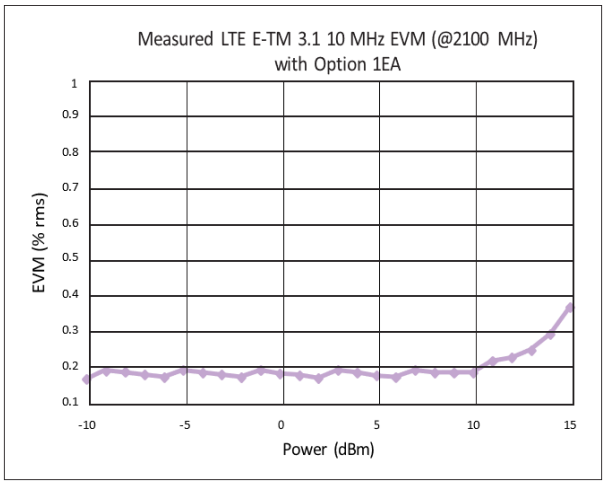
1. Performance evaluated at bottom, middle, and top of bands shown.
2. This is rms power. Convert from rms to peak envelope power (PEP) with the following equation: PEP = rms power + crest factor (for example: 3GPP test model 1 with 64 DPCH has a crest factor > 11 dB, therefore at +5 dBm rms the PEP = 5 dBm + 11 dB = +16 dBm PEP).
3. Measurement configuration: reference channel integration BW: 9.5 MHz, offset channel integration BW: 9 MHz, channel offset: 10 MHz.
4. 802.16e WiMAX signal configuration–bandwidth: 10 MHz, FFT: 1024, frame length: 5 ms, guard period: 1/ 8, symbol rolloff: 5%, content: 30 symbols of PN9 data.

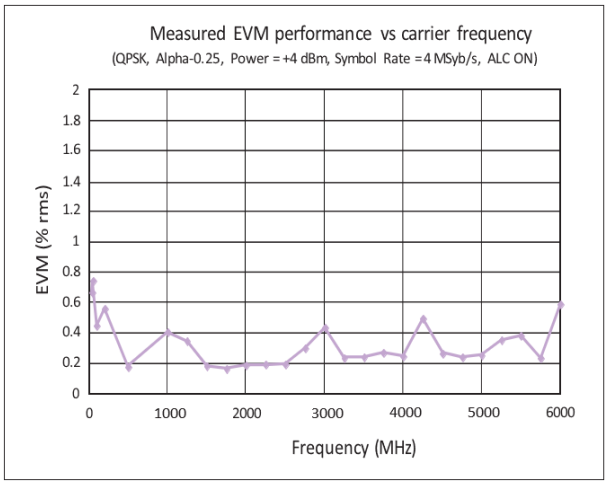
EVM performance data [1, 2]										
Format	**GSM**		**EDGE**		**cdma2000/IS95A**		**W-CDMA**		**LTE FDD [3]**	
Modulation type	GMSK (bursted)		3pi/8 8PSK (bursted)		QPSK		QPSK		64 QAM	
Modulation rate	270.833 ksps		70.833 ksps		1.2288 Mcps		3.84 Mcps		10 MHz BW	
Configuration	1 timeslot		1 timeslot		Pilot channel		1 DPCH		E-TM 3.1	
Frequency [4]	800 to 900 MHz 1800 to1900 MHz		800 to 900 MHz 1800 to 1900 MHz		800 to 900 MHz 1800 to 1900 MHz		1800 to 2200 MHz		1800 to 2200 MHz	
EVM power level	≤ 7 dBm		≤ 7 dBm		≤ 7 dBm		≤ 7 dBm		≤ 7 dBm	
EVM power level with Option 1EA	≤ 13 dBm		≤ 13 dBm		≤ 13 dBm		≤ 13 dBm		≤ 13 dBm	
EVM/global phase error	**Spec**	**Typ**	**Spec**	**Typ**	**Spec**	**Typ**	**Spec**	**Typ**	**Measured**	
	rms 0.8 °	0.2 °	1.2%	0.75%	1.3%	0.8%	1.2%	0.8%	0.2%	
Format	**802.11a/g**	**802.11ac[5]**	**QPSK**				**16 QAM**			
Modulation type	64 QAM	256 QAM	QPSK				16 QAM			
Modulation rate	54 Mbps	80 MHz	4 Msps (root-Nyquist filter α = 0.25)							
Frequency [4]	2400 to 2484 MHz	5.775 GHz	≤ 3 GHz		≤ 6 GHz		≤ 3 GHz		≤ 6 GHz	
	5150 to 5825 MHz									
EVM power level	≤ –5 dBm	≤ –5 dBm	≤ 4 dBm		≤ 4 dBm		≤ 4 dBm		≤ 4 dBm	
EVM power level with Option 1EA	≤ 2 dBm	≤ 2 dBm	≤ 10 dBm		≤ 10 dBm		≤ 10 dBm		≤ 10 dBm	
EVM	**Measured**	**Measured**	**Spec**	**Typ**	**Spec**	**Typ**	**Spec**	**Typ**	**Spec**	**Typ**
	0.3%	0.4%	1.2%	0.8%	1.9%	1.1%	1.1%	0.65%	1.5%	0.9%

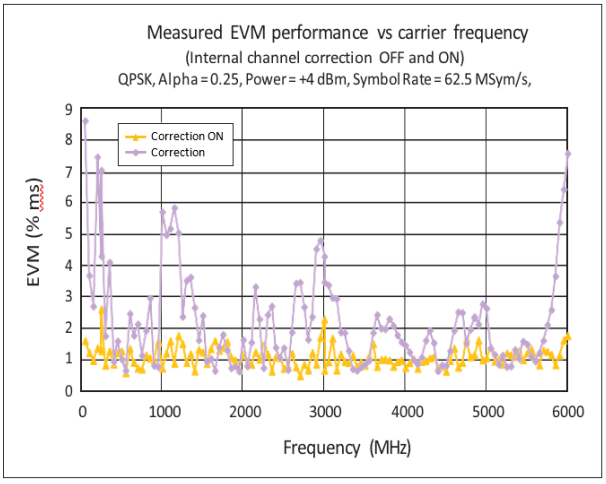
1. EVM specifications apply for the default ARB file setup conditions with the default ARB files supplied with the instrument.
2. EVM specifications apply after execution of I/Q calibration when the instrument is maintained within ± 5 °C of the calibration temperature.
3. LTE FDD E-TM 3.1, 10 MHz, 64 QAM PDSCH, full resource block. Measured EVM after DC calibration.
4. Performance evaluated at bottom, middle, and top of bands shown.
   WLAN 802.11ac 80 MHz, 256 QAM, MCS 8, 7 symbols, no filtering. Channel corrections enabled. Rx equalizer training preamble only.

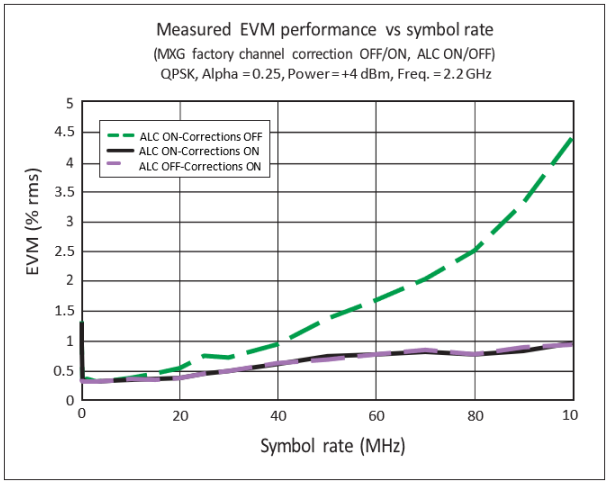

Measured EVM performance vs. power
802.11ac 80 MHz 256 QAM @ 5.775 GHz
Rx equalizer training preamble only
(MXG factory channel corrections enabled)
EVM (% rms)
Power level (dBm)


Measured EVM performance vs. power
802.11ac 160 MHz 256 QAM @ 5.25 GHz
Rx equalizer training preamble only
(MXG factory channel corrections enabled)
EVM (% rms)
Power level (dBm)


Measured LTE E-TM 3.1 10 MHz EVM (@2100 MHz)
with Option 1EA
EVM (% rms)
Power (dBm)


Measured EVM performance vs carrier frequency
(QPSK, Alpha-0.25, Power = +4 dBm, Symbol Rate = 4 MSyb/s, ALC ON)
EVM (% rms)
Frequency (MHz)


Measured EVM performance vs carrier frequency
(Internal channel correction OFF and ON)
QPSK, Alpha = 0.25, Power = +4 dBm, Symbol Rate = 62.5 MSym/s,
Correction ON
Correction
EVM (% rms)
Frequency (MHz)


Measured EVM performance vs symbol rate
(MXG factory channel correction OFF/ON, ALC ON/OFF)
QPSK, Alpha = 0.25, Power = +4 dBm, Freq. = 2.2 GHz
ALC ON-Corrections OFF
ALC ON-Corrections ON
ALC OFF-Corrections ON
EVM (% rms)
Symbol rate (MHz)

Bit error rate [BER] analyzer (Option UN7)	
Clock rate	100 Hz to 60 MHz (usable to 90 MHz)
Data patterns	PN9, 11, 15, 20, 23
Resolution	10 digits
Bit sequence length	100 bits to 4,294 Gbits after synchronization
Other features	Input clock phase adjustment and gate delay Direct measurement triggering Data and reference signal outputs Real-time display Bit count Error-bit-count Bit error rate Pass/fail indication Valid data and clock detection Automatic re-synchronization Special pattern ignore

General Specifications

Remote programming	
Interfaces	GPIB IEEE-488.2, 1987 with listen and talk LAN 1000BaseT LAN interface, LXI class C compliant USB Version 2.0
Control languages	Control languages SCPI Version 1997.0
Compatibility languages	Keysight Technologies: N5181A\61A, N 5182A\62A, N5183A, E4438C, E4428C, E442xB, E443xB, E8241A, E8244A, E8251A, E8254A, E8247C, E8257C/D, E8267C/D, 8648 Series, 8656B, E8663B, 8657A/B, 8662A, 8663A Aeroflex Incorporated: 3410 Series Rohde & Schwarz: SMB100A, SMBV100A, SMU200A, SMJ100A, SMATE200A, SMIQ, SML, SMV
Power requirements	
100/120 VAC, 50/60/400 Hz 220/240 VAC, 50/60 Hz 160 W maximum (N5181B) 300 W maximum (N5182B)	
Operating temperature range	
0 to 55 °C	
Storage temperature range	
-40 to 70 °C	
Operating and storage altitude	
Up to 4,600 meters Up to 3,000 meters (Option 660 only)	
Indoor use	
For indoor use only	
Humidity	
Maximum Relative Humidity (non-condensing): 95%RH up to 40 °C, decreases linearly to 45%RH at 55 °C.	
Environmental stress	
Samples of this product have been type tested in accordance with the Keysight Environmental Test Manual and verified to be robust against the environmental stresses of storage, transportation and end-use; those stresses include but are not limited to temperature, humidity, shock, vibration, altitude, and power line conditions; test methods are aligned with IEC 60068-2 and levels are similar to MIL-PRF-28800F Class 3	

1. From 40 °C to 55 °C, the maximum % Relative Humidity follows the line of constant dew point.

Safety		
Complies with European Low Voltage Directive 2006/95/EC		
– IEC/EN 61010-1, 2nd Edition – Canada: CSA C22.2 No. 61010-1 – USA: UL std no. 61010-1, 2nd Edition – German Acoustic statement	Acoustic noise emission LpA < 70 dB Operator position Normal position Per ISO 7779	Geraeuschemission LpA < 70 dB Am Arbeitsplatz Normaler Betrieb Nach DIN 45635 t.19
Complies with European EMC Directive 2004/108/EC		
– IEC/EN 61326-1or IEC/EN 61326-2-1 CISPR Pub 11 Group 1, class A AS/NZS CISPR 11 ICES/NMB-001	This ISM device complies with Canadian ICES-001; cet appareil ISM est conforme a la norme NMB-001 du Canada	
Memory		
– Memory is shared by instrument states, user data files, sweep list files, waveform sequences, and other files 3 GB (30 GB with Option 009) memory available in the N5182B Security Option 006 allows storage of up to 8 GB on SD card Depending on how the memory is utilized, a maximum of 1000 instrument states can be saved		
Security (Option 006)		
No internal non-volatile memory (Option SD0) Disable/remove any internal non-volatile memory or solid state drive User will not be able to store any files in the internal memory of the instrument Not compatible with instrument hardware option 009 (Internal Solid State Memory) and option 660 (Base Band Generator with Real-Time Capability) Requires firmware B.01.80 or newer		
Self-test		
Internal diagnostic routines test most modules in a preset condition; for each module, if its node voltages are within acceptable limits, the module passes the test.		
Weight		
N5181B: ≤ 13.6 kg (30 lb) net, ≤ 28.6 kg (63 lb) shipping N5182B: ≤ 15.9 kg (35 lb) net, ≤ 30.8 kg (68 lb) shipping		
Dimensions		
88 mm Hx 426 mm W x 489 mm L (length includes rear panel feet) (3.5 in H x 16.8 in W x 19.2 in L) Max length (L) include RF connector tip to end of rear panel feet is 508 mm (20 in)		
Recommended calibration cycle		
36 months		
ISO compliant		
This instrument is manufactured in an ISO-9001 registered facility in concurrence with Keysight Technologies' commitment to quality.		

Inputs and Outputs

Front panel connectors	
RF output	Outputs the RF signal via a precision N type female connector; see output section for reverse power protection information
I and Q inputs	BNC input accepts “in-phase” and “quadrature” input signals for I/Q modulation; nominal input imped- ance is 50 Ω, damage levels are 1 Vrms and 5 Vpeak
USB 2.0	Used with a memory stick for transferring instrument states, licenses and other files into or out of the instrument; also used with U2000, U848X and U202X Series USB power sensors.
Rear panel connectors	
Rear panel inputs and outputs are 3.3 V CMOS, unless indicated otherwise; CMOS inputs will accept 5 V CMOS, 3 V CMOS, or TTL voltage levels	
RF output (Option 1EM)	Outputs the RF signal via a precision N type female connector
I and Q inputs (Option 1EM)	Accepts “in-phase” and “quadrature” input signals for I/Q modulation SMB connector, nominal input impedance is 50 Ω; damage levels are 1 Vrms and 5 Vpeak; Option 1EM units will come with 2 SMB to BNC adapters
I and Q outputs	BNC outputs the analog I/Q modulation signals from the internal baseband generator; nominal output impedance 50 Ω, DC coupled; damage levels ± 2 V
I bar and Q bar outputs (Option 1EL)	BNC outputs the complement of the I and Q signals for differential applications
Event 1	This connector outputs the programmable timing signal generated by marker 1 The marker signal can also be routed internally to control the RF blanking and ALC hold functions; this signal is also available on the AUX I/O connector With bit error rate analyzer (Option UN7) this connector is used for data input Damage levels are > +8 V and < –4 V
Pattern trigger	Accepts signal to trigger internal pattern generator to start single pattern output, for use with the internal baseband generators Accepts CMOS signal with minimum pulse width of 10 ns Damage levels are > +8 V and < –4 V
BBTRIG 1	For arbitrary and real-time baseband generators I/O such as Markers or trigger inputs With bit error rate analyzer (Option UN7) this connector is used for clock input
BBTRIG 2	For arbitrary and real-time baseband generators I/O such as Markers or trigger inputs With bit error rate analyzer (Option UN7) this connector is used for gate input
Sweep out	Generates output voltage, 0 to +10 V when the signal generator is sweeping; this output can also be programmed to indicate when the source is settled or output pulse video and is TTL and CMOS compatible in this mode; output impedance < 1 Ω, can drive 2 kΩ; damage levels are ± 15 V.
Ext 1	External AM/FM/PM #1 input; nominal input impedance is 50 Ω/600 Ω/1M Ω, nominal; damage levels are ± 5 V
Ext 2	External AM/FM/PM #2 input; nominal input impedance is 50 Ω/600 Ω /1M Ω, nominal; damage levels are ± 5 V
LF OUT	0 to 5 V peak into 50 Ω, –5 V to 5 V offset, nominal
Pulse	External pulse modulation input; this input is TTL and CMOS compatible; low logic levels are 0 V and high logic levels are +1 V; nominal input impedance is 50 Ω; input damage levels are ≤ –0.3 V and ≥ +5.3 V.
Trigger in	Accepts TTL and CMOS level signals for triggering point-to-point in sweep mode; damage levels are ≤ –0.3 V and ≥ +5.3 V.

Trigger out	Outputs a TTL and CMOS compatible level signal for use with sweep mode. The signal is high at start of dwell, or when waiting for point trigger in manual sweep mode, and low when dwell is over or point trigger is received. This output can also be programmed to indicate when the source is settled, pulse synchronization, or pulse video. Nominal output impedance 50 Ω Input damage levels are ≤ –0.3 V and ≥ +5.3 V
Reference input	Accepts a 10 MHz reference signal used to frequency lock the internal timebase; Option 1ER adds the capability to lock to a frequency from 1 MHz to 50 MHz; nominal input level –3 to +20 dBm, impedance 50 Ω, sine or square waveform.
10 MHz out	Outputs the 10 MHz reference signal used by internal timebase; level nominally +3.9 dBm; nominal output impedance 50 Ω; input damage level is +16 dBm.
LO in (Option 012)	Accepts a signal from a primary signal generator that is used as the LO for MXG vector in order to configure a phase coherent system; nominal input levels between 0 to +12 dBm; nominal input impedance 50 Ω.
LO out (Option 012)	Outputs a reference signal that can be used in a phase coherent system; nominal output levels between 0 to +12 dBm; nominal output impedance 50 Ω.
DAC Clk In (Option 012)	Reserved for future use.
Digital bus I/O	To be used with PXB or N5102A digital signal interface module.
Aux I/O	Aux I/O port sends and/or receives auxiliary signaling information: For Option UN7 this connector is used to output reference data, clock, error signals, and more. Output markers to an external device from arbitrary waveform or real-time generation application such as: frame markers, pulse-per-second, even-second, and more. Input signals from external DUT to modify characteristics of a signal being generated such as changing output power (power control loop testing), advancing or delaying timing (timing advance loop testing), HARQ ACK/NAK delivery (HARQ process loop testing) or streaming external data, clock and symbol synch for custom modulation. I/O is application specific (CDMA, 3GPP, GNSS, LTE, custom). See User Guide or Signal Studio help for more details. Connector type: 36 pin 3M connector (part number N10236-52B2PC). The mating connector is a 3M 10136-3000 wire mount plug or 3M 10136-8000 IDC plug with a 3M 10336 shell. For Option N5180431B real-time custom modulation the following pin numbers are assigned: Data input = pin 23 Data clock input = pin 29 Symbol sync input = pin 25 Burst input = pin 27 Data output = pin 35 Data clock output = pin 6 Symbol sync output = pin 37 Event 1 output = pin 1 Event 2 output = pin 33
USB 2.0	The USB connector provides remote programming functions via SCPI
LAN (1000 BaseT)	The LAN connector provides the same SCPI remote programming functionality as the GPIB connector and is also used to access the internal Web server and FTP server Supports DHCP, sockets SCPI, VXI-11 SCPI, connection monitoring, dynamic hostname services, TCP keep alive LXI class C compliant Trigger response time for the immediate LAN trigger is 0.5 ms (minimum), 4 ms (maximum), 2 ms (typical); delayed/ alarm trigger is unknown Trigger output response time is 0.5 ms (minimum), 4 ms (maximum), 2 ms (typical)
GPIB	The GPIB connector provides remote programming functionality via SCPI

Related Literature

Keysight X-Series Signal Generators	
MXG Configuration Guide	5990-9959EN
EXG Data Sheet	5991-0039EN
EXG Configuration Guide	5990-9958EN
X-Series Signal Generator Brochure	5990-9957EN
Signal Studio Software Brochure	5989-6448EN
N5182BX07 User Guide	N5182-90001

Confidently Covered by Keysight Services

Prevent delays caused by technical questions, or system downtime due to instrument maintenance and repairs with Keysight Services. Keysight Services are here to support your test needs with expert technical support, instrument repair and calibration, software support, training, alternative acquisition program options, and more.

A KeysightCare agreement provides dedicated, proactive support through a single point of contact for instruments, software, and solutions. KeysightCare covers an extensive group of instruments, application software, and solutions and ensures optimal uptime, faster response, faster access to experts, and faster resolution.

Keysight Services

Offering	1. Benefits
KeysightCare KEYSIGHTCARE	KeysightCare provides elevated support for Keysight instruments and software, with access to technical support experts that respond within a specified time and ensure committed repair and calibration turnaround times (TAT). KeysightCare offers multiple service agreement tiers, including KeysightCare Assured, Enhanced, and Application Software Support. See the KeysightCare data sheet for details.
KeysightCare Assured	KeysightCare Assured goes beyond basic warranty with repair services that include committed TAT and unlimited access to technical experts.
KeysightCare Enhanced	KeysightCare Enhanced includes all the benefits of KeysightCare Assured plus Keysight's accurate and reliable calibration services, accelerated, and committed TAT, and technical response.
Keysight Support Portal & Knowledge Center	All KeysightCare tiers include access to the Keysight Support Portal where you can manage support and service resources related to your assets such as service requests, and status, or browse the Knowledge Center.
Education Services	Build confidence and gain new skills to make accurate measurements, with flexible Education Services developed by Keysight experts. Including Start-up Assistance.
Alternative product acquisition	
KeysightAccess	Reduce budget challenges with a subscription service enabling you to get the instruments, software, and technical support you want for your test needs.

Recommended Services

Maximize your test system up-time by securing technical support, repair, and calibration services with committed response and turnaround times. 1-year KeysightCare Assured is included in every new instrument purchase Obtain multi-year KeysightCare upfront to eliminate the need for lengthy and tedious paperwork and yearly requests for maintenance budget. Plus, you benefit from secured service for 2, 3, or 5 years.

Service	Function
KeysightCare Enhanced*	**Includes Tech Support, Warranty and Calibration**
R-55B-001-1	KeysightCare Enhanced – Upgrade 1 year
R-55B-001-2	KeysightCare Enhanced – Extend to 2 years
R-55B-001-3	KeysightCare Enhanced – Extend to 3 years (Recommended)
R-55B-001-5	KeysightCare Enhanced – Extend to 5 years (Recommended)
KeysightCare Assured	**Includes Tech Support and Warranty**
R-55A-001-2	KeysightCare Assured – Extend to 2 years
R-55A-001-3	KeysightCare Assured – Extend to 3 years
R-55A-001-5	KeysightCare Assured – Extend to 5 years
Start-Up Assistance	
PS-S10	Included – instrument fundamentals and operations starter
PS-S20	Optional, technology & measurement science standard learning

* Available in select countries. For details, please view the datasheet. R-55B-001-2/3/5 must be ordered with R-55B-001-1.

Learn more at: www.keysight.com

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